

Q1 - 24 January - Shift 1

$$\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right) \text{ is equal to}$$

Space for your notes:

(1) $\frac{\pi}{4}$

(2) $\frac{\pi}{2}$

(3) $\frac{\pi}{3}$

(4) $\frac{\pi}{6}$

Q2 - 25 January - Shift 1

If the sum of all the solutions of $\tan^{-1}\left(\frac{2x}{1-x^2}\right) + \cot^{-1}\left(\frac{1-x^2}{2x}\right) = \frac{\pi}{3}$,

Space for your notes:

$-1 < x < 1, x \neq 0$, is $\alpha - \frac{4}{\sqrt{3}}$, then α is equal to

Q3 - 30 January - Shift 2

Let $a_1 = 1, a_2, a_3, a_4, \dots$ be consecutive natural numbers. Then $\tan^{-1}\left(\frac{1}{1+a_1a_2}\right) + \tan^{-1}\left(\frac{1}{1+a_2a_3}\right)$

Space for your notes:

$+ \dots + \tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right)$ is equal to

(1) $\frac{\pi}{4} - \cot^{-1}(2022)$

(2) $\cot^{-1}(2022) - \frac{\pi}{4}$

(3) $\tan^{-1}(2022) - \frac{\pi}{4}$

(4) $\frac{\pi}{4} - \tan^{-1}(2022)$

Q4 - 31 January - Shift 1

Questions with Solutions

MathonGo

If $\sin^{-1} \frac{\alpha}{17} + \cos^{-1} \frac{\alpha}{5} - \tan^{-1} \frac{\alpha}{36} = 0$, $0 < \alpha < 13$,

Space for your notes:

then $\sin^{-1}(\sin \alpha) + \cos^{-1}(\cos \alpha)$ is equal to

- (1) π
- (2) 16
- (3) 0
- (4) $16 - 5\pi$

Q5 - 31 January - Shift 1

If the sum and product of four positive consecutive terms of a G.P., are 126 and 1296, respectively, then the sum of common ratios of all such GPs is

Space for your notes:

- (1) 7
- (2) $\frac{9}{2}$
- (3) 3
- (4) 14

Q6 - 31 January - Shift 2

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Questions with Solutions

MathonGo

Let $(a, b) \subset (0, 2\pi)$ be the largest interval for which

$$\sin^{-1}(\sin \theta) - \cos^{-1}(\sin \theta) > 0, \theta \in (0, 2\pi),$$

holds. If

$$\alpha x^2 + \beta x + \sin^{-1}(x^2 - 6x + 10) + \cos^{-1}(x^2 - 6x + 10) = 0$$

and $\alpha - \beta = b - a$, then α is equal to :

(1) $\frac{\pi}{48}$

(2) $\frac{\pi}{16}$

(3) $\frac{\pi}{8}$

(4) $\frac{\pi}{12}$

Space for your notes:

Q7 - 01 February - Shift 1

Let S be the set of all solutions of the equation

$$\cos^{-1}(2x) - 2\cos^{-1}(\sqrt{1-x^2}) = \pi, x \in \left[-\frac{1}{2}, \frac{1}{2}\right].$$

Then $\sum_{x \in S} 2\sin^{-1}(x^2 - 1)$ is equal to

(1) 0

(2) $-\frac{2\pi}{3}$

(3) $\pi - \sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$

(4) $\pi - 2\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$

Space for your notes:

Q8 - 01 February - Shift 2

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Let $S = \left\{ x \in \mathbb{R} : 0 < x < 1 \text{ and } 2 \tan^{-1} \left(\frac{1-x}{1+x} \right) = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \right\}$. If

Space for your notes:

$n(S)$ denotes the number of elements in S then :

(1) $n(S) = 2$ and only one element in S is less than $\frac{1}{2}$.

(2) $n(S) = 1$ and the element in S is more than $\frac{1}{2}$.

(3) $n(S) = 1$ and the element in S is less than $\frac{1}{2}$.

(4) $n(S) = 0$

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (3)**Q2 (2)****Q3 (3)****Q4 (1)****Q5 (1)****Q6 (4)****Q7 (2)****Q8 (3)**

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Q1 (3)

$$\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$$

$$= \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) + \sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \frac{\pi}{3}$$

Q2 (2)

Case I : $x > 0$

$$\tan^{-1}\frac{2x}{1-x^2} + \tan^{-1}\frac{2x}{1-x^2} = \frac{\pi}{3}$$

$$x = 2 - \sqrt{3}$$

Case II : $x < 0$

$$\tan^{-1}\frac{2x}{1-x^2} + \tan^{-1}\frac{2x}{1-x^2} + \pi = \frac{\pi}{3}$$

$$x = \frac{-1}{\sqrt{3}} \Rightarrow \alpha = 2$$

Q3 (3)

$$a_2 - a_1 = a_3 - a_2 = \dots = a_{2022} - a_{2021} = 1.$$

$$\therefore \tan^{-1}\left(\frac{a_2 - a_1}{1 + a_1 a_2}\right) + \tan^{-1}\left(\frac{a_3 - a_2}{1 + a_2 a_3}\right) + \dots + \tan^{-1}\left(\frac{a_{2022} - a_{2021}}{1 + a_{2021} a_{2022}}\right)$$

$$= \left[\left(\tan^{-1} a_2 \right) - \tan^{-1} a_1 \right] + \left[\tan^{-1} a_3 - \tan^{-1} a_2 \right] + \dots + \left[\tan^{-1} a_{2022} - \tan^{-1} a_{2021} \right]$$

$$= \tan^{-1} a_{2022} - \tan^{-1} a_1$$

$$= \tan^{-1}(2022) - \tan^{-1} 1 = \tan^{-1} 2022 - \frac{\pi}{4} \text{ (option 3)}$$

$$= \left(\frac{\pi}{2} - \cot^{-1}(2022) \right) - \frac{\pi}{4}$$

$$= \frac{\pi}{4} - \cot^{-1}(2022) \text{ (option 1)}$$

Q4 (1)

$$\cos^{-1} \frac{4}{5} = \tan^{-1} \frac{3}{4}$$

$$\therefore \sin^{-1} \frac{\alpha}{17} = \tan^{-1} \frac{77}{36} - \tan^{-1} \frac{3}{4} = \tan^{-1} \left(\frac{\frac{77}{36} - \frac{3}{4}}{1 + \frac{77}{36} \cdot \frac{3}{4}} \right)$$

$$\sin^{-1} \frac{\alpha}{17} = \tan^{-1} \frac{8}{15} = \sin^{-1} \frac{8}{17}$$

$$\Rightarrow \frac{\alpha}{17} = \frac{8}{17} \Rightarrow \alpha = 8$$

$$\therefore \sin^{-1}(\sin 8) + \cos^{-1}(\cos 8)$$

$$= 3\pi - 8 + 8 - 2\pi$$

$$= \pi$$

Q5 (1)

$$a, ar, ar^2, ar^3 \quad (a, r > 0)$$

$$ar^6 = 1296$$

$$a^2 r^3 = 36$$

$$a = \frac{6}{r^{3/2}}$$

$$a + ar + ar^2 + ar^3 = 126$$

$$\frac{1}{r^{3/2}} + \frac{r}{r^{3/2}} + \frac{r^2}{r^{3/2}} + \frac{r^3}{r^{3/2}} = \frac{126}{6} = 21$$

$$(r^{-3/2} + r^{3/2}) + (r^{1/2} + r^{-1/2}) = 21$$

$$r^{1/2} + r^{-1/2} = A$$

$$r^{-3/2} + r^{3/2} + 3A = A^3$$

$$A^3 - 3A + A = 21$$

$$A^3 - 2A = 21$$

$$A = 3$$

$$\sqrt{r} + \frac{1}{\sqrt{r}} = 3$$

$$r + 1 = 3\sqrt{r}$$

$$r^2 + 2r + 1 = 9r$$

$$r^2 - 7r + 1 = 0$$

Q6 (4)

$$\sin^{-1} \sin \theta - \left(\frac{\pi}{2} - \sin^{-1} \sin \theta \right) > 0$$

$$\Rightarrow \sin^{-1} \sin \theta > \frac{\pi}{4}$$

$$\Rightarrow \sin \theta > \frac{1}{\sqrt{2}}$$

$$\text{So, } \theta \in \left(\frac{\pi}{4}, \frac{3\pi}{4} \right)$$

$$\theta \in \left(\frac{\pi}{4}, \frac{3\pi}{4} \right) = (a, b)$$

$$b - a = \frac{\pi}{2} = \alpha - \beta$$

$$\Rightarrow \beta = \alpha - \frac{\pi}{2}$$

$$\Rightarrow \alpha x^2 + \beta x + \sin^{-1} \left[(x-3)^2 + 1 \right] + \cos^{-1} \left[(x-3)^2 + 1 \right] = 0$$

$$x = 3, 9\alpha + 3\beta + \frac{\pi}{2} + 0 = 0$$

$$\Rightarrow 9\alpha + 3 \left(\alpha - \frac{\pi}{2} \right) + \frac{\pi}{2} = 0$$

$$\Rightarrow 12\alpha - \pi = 0$$

$$\alpha = \frac{\pi}{12}$$

Q7 (2)

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$$\cos^{-1}(2x) - 2\cos^{-1}\sqrt{1-x^2} = \pi$$

$$\cos^{-1}(2x) - \cos^{-1}(2(1-x^2) - 1) = \pi$$

$$\cos^{-1}(2x) - \cos^{-1}(1-2x^2) = \pi$$

$$-\cos^{-1}(1-2x^2) = \pi - \cos^{-1}(2x)$$

Taking cos both sides we get

$$\cos(-\cos^{-1}(1-2x^2)) = \cos(\pi - \cos^{-1}(2x))$$

$$1-2x^2 = -2x$$

$$2x^2 - 2x - 1 = 0$$

$$\text{On solving, } x = \frac{1-\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}$$

$$\text{As } x \in [-1/2, 1/2], x = \frac{1+\sqrt{3}}{2} \text{ is rejected}$$

$$\text{So } x = \frac{1-\sqrt{3}}{2} \Rightarrow x^2 - 1 = -\sqrt{3}/2$$

$$= 2\sin^{-1}(x^2 - 1) = 2\sin^{-1}\left(\frac{-\sqrt{3}}{2}\right) = \frac{-2\pi}{3}$$

Q8 (3)

$$0 < x < 1$$

$$2 \tan^{-1} \left(\frac{1-x}{1+x} \right) = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$$

$$\tan^{-1} x = \theta \in \left(0, \frac{\pi}{4} \right) \therefore x = \tan \theta$$

$$2 \tan^{-1} \left(\tan \left(\frac{\pi}{4} - \theta \right) \right) = \cos^{-1} (\cos 2\theta)$$

$$2 \left(\frac{\pi}{4} - \theta \right) = 2\theta \therefore 4\theta = \frac{\pi}{2} \therefore \theta = \frac{\pi}{8}$$

$$x = \tan \frac{\pi}{8} \therefore x = \sqrt{2} - 1 \approx 0.414$$